

The Art of Designing a Custom Library for Arduino IDE

This article is about how to create an Arduino library and the opportunities in open source programming with the Arduino IDE. It will also show you how to interface a temperature sensor LM35 using your own library file.

Before we start speaking about this really cool thing called a custom library design, let us first discuss a few fascinating ideas related to open source programming and the basics of the Arduino IDE. The article assumes that the readers have a basic knowledge of electronics, and are aware of Arduino and the related sensors.

Open source programming and hardware

A few parameters for a programming language to be declared as open source are:

- The source code must be open and accessible
- It should be freely redistributable, yet the integrity of the author's source code should be maintained
- Its licence must not restrict other software
- No discrimination against fields of endeavour

Many large institutions have sprung up to support the development of the open source movement, including the Apache Software Foundation, which supports Apache Hadoop and the open source HTTP server, Apache HTTP. The open source initiative is not only limited to engineering but extends to non-engineering areas too, including medicine, law and order, fashion design, etc. And the concept of 'openness' now encompasses hardware too.

A few examples of open source hardware initiatives are:

- Openmoko — a family of open source mobile phones, including the hardware specification and the operating system.
- Arduino — a microcontroller platform for embedded system designers.
- GizmoSphere — an open source development platform for the embedded design community.
- Simputer — an open hardware handheld computer, designed in India.
- LEON: A family of open source microprocessors distributed in a library with peripheral IP cores, open SPARC V8 specifications, and implementable under the GNU GPL.
- Tinkerforge — a system of open source stackable microcontroller building blocks.

LM35 temperature sensor

The LM35 is a precision integrated circuit (IC) temperature sensor whose output voltage is directly proportional to the temperature (10 mV/°C). The LM35 generates a higher output voltage than thermocouples and may not require the output

voltage to be amplified. It does not require any external calibration or trimming, and maintains an accuracy of $\pm 0.4^{\circ}\text{C}$ at room temperature and $\pm 0.8^{\circ}\text{C}$ over a range of 0°C to $+100^{\circ}\text{C}$. The scale factor is $.01\text{V}/^{\circ}\text{C}$; it draws only 60 micro amps and possesses a low self-heating capability.

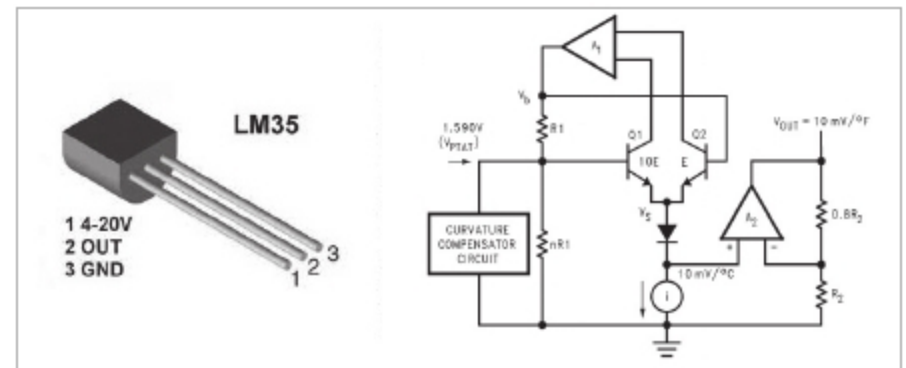


Figure 1: Pin and block diagrams of LM35

There are two transistors in the block diagram above — one has ten times the emitter area of the other. This means it has one-tenth of the current density, since the same current is going through both transistors. This causes a voltage across the resistor R1 that is proportional to the absolute temperature, and is almost linear across the range. The amplifier A1 ensures that the voltage at the base of the left transistor (Q1) is proportional to the absolute temperature by comparing the output of the two transistors. The amplifier A2 converts absolute temperature in Kelvin into either Fahrenheit (LM34) or Celsius (LM35). The 'i' in the block diagram is a constant current source circuit. The resistors R1 and R2 are calibrated in the factory to produce a highly accurate temperature sensor.

Arduino IDE

Arduino is an open hardware development board that can be used to design and build devices for use in everyday life. It is a physical computing platform based on a simple I/O board. A development environment can be used to develop standalone interactive objects or can be connected to software running on a computer. An Arduino board consists of an Atmel AVR microcontroller (ATmega328, ATmega168 and ATmega8 versions) and supporting hardware modules to facilitate programming and incorporate it into other circuits. One of the main features of this IDE is that the microcontroller is pre-programmed with a bootloader, so an external programmer is not necessary.